

Box Class Reference

The Box class represents an axis-aligned box in 3D space (usually a bounding box). [More...](#)

```
#include <box.h>
```

Public Member Functions

Box (const Point &point= Point ())
Box (const Point &minimum, const Point &maximum)
void expand (const Point &p)
void expand (const Box &b)
Point center () const
float radius () const
Point min () const
Point max () const

Detailed Description

The **Box** class represents a box in 3D space through its (min, max) corners. This class is suitable for representing axis-aligned boxes (but not oriented boxes). A typical use is for representing the bounding box of an object or group of objects.

Constructor & Destructor Documentation

```
Box::Box ( const Point & point = Point() )
```

Constructs a box containing a single point (minimum=maximum=point).

```
Box::Box ( const Point & minimum,
           const Point & maximum
         )
```

Constructs a box from two points (minimum, maximum).

Member Function Documentation

Point Box::center () const

Draws the box (wireframe).

Returns the center of the box.

void Box::expand (const Point & p)

Expands the box so that it includes point p.

void Box::expand (const Box & b)

Expands the box so that it includes box b.

Point Box::max () const

Returns the maximum corner of the box.

Point Box::min () const

Returns the minimum corner of the box.

float Box::radius () const

Returns half the box's diagonal, that is, the radius of a sphere bounding the box.